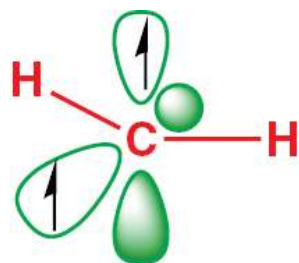
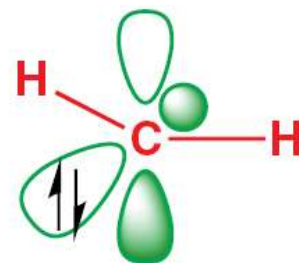


Chapter 9

Synthesis and Reactions of Carbenes and Nitrenes



triplet carbene



singlet carbene





Contents

9.1 Photolysis of diazomethane produces a carbene

9.2 Ways to make carbenes

9.3 Carbenes can be divided into two types

9.4 Carbenes react with alkenes to give cyclopropanes

9.5 Insertion into C–H bonds

9.6 Rearrangement reactions

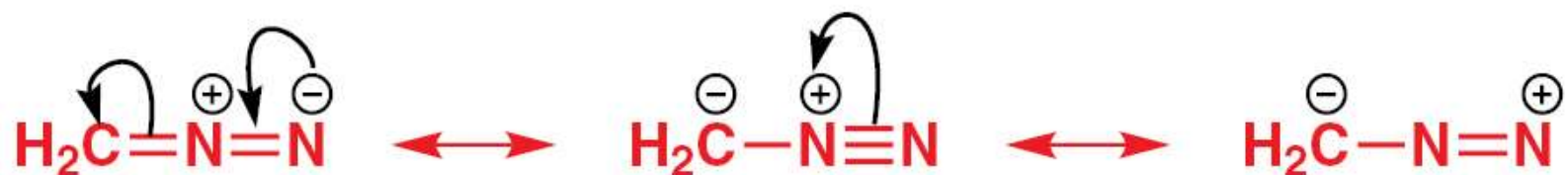
9.7 Nitrenes are the nitrogen analogues of carbenes

9.8 Alkene metathesis

9.9 Summary **References: Chapter 38 of Clayden's Organic Chemistry**

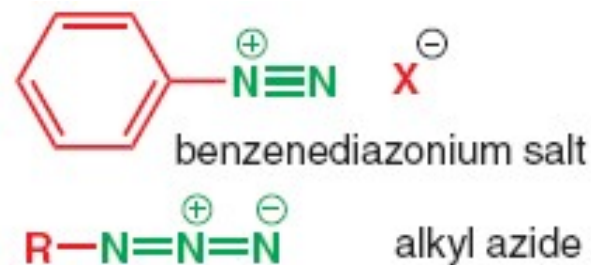
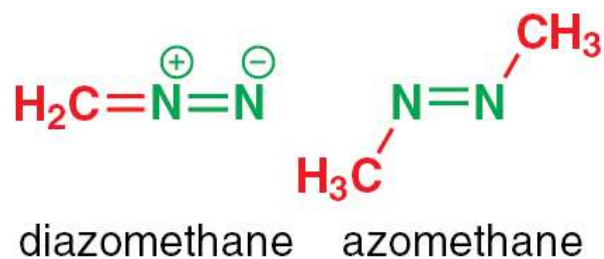


Diazomethane (重氮甲烷)



Concentrated solutions of diazomethane are **explosive**.

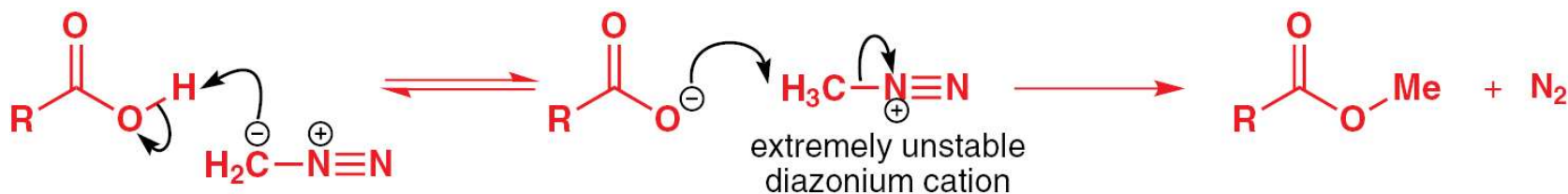
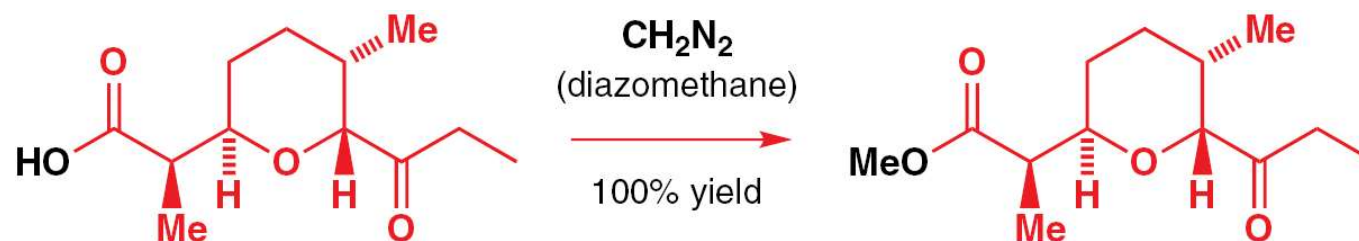
常见的含有连续N-N键的化合物



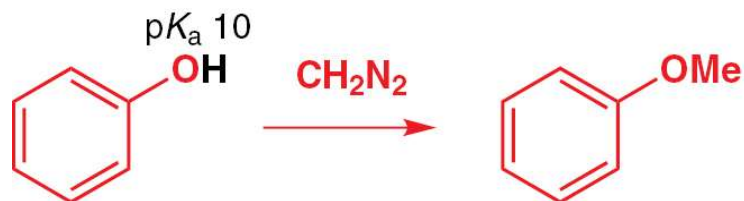


重氮甲烷的用途

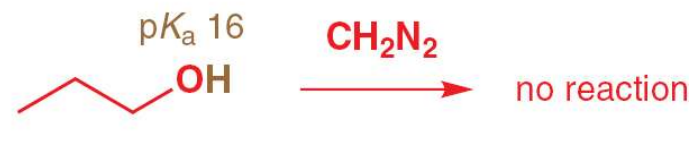
酯化反应



苯酚醚化反应



酸性足以使重氮甲烷质子化



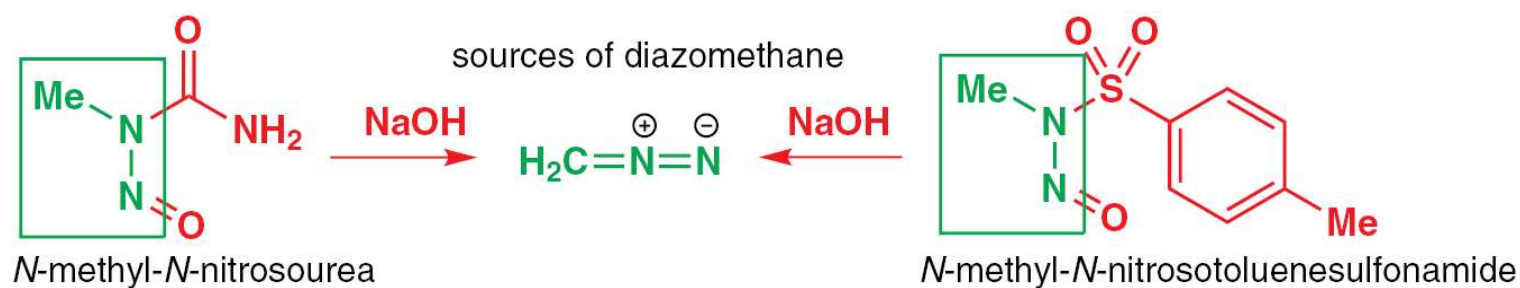
不和醇类反应



Preparation of diazomethane

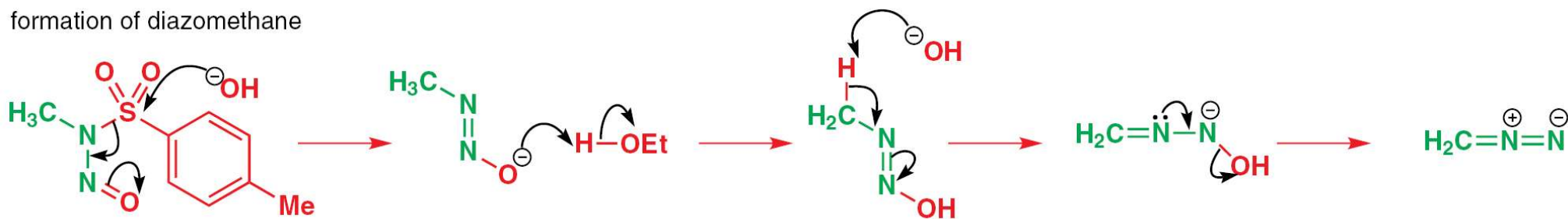
(了解即可)

重氮甲烷制备方法:



反应机理:

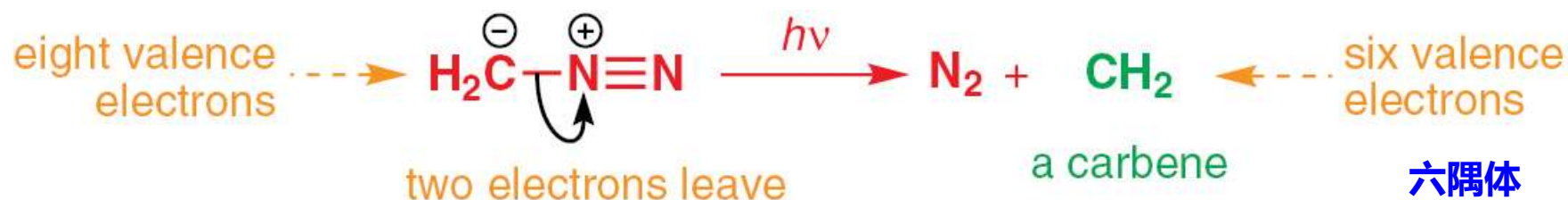
formation of diazomethane



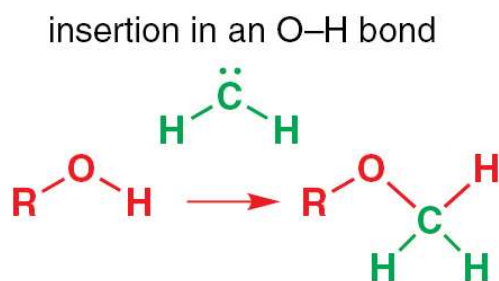


9.1 Photolysis of diazomethane produces a carbene

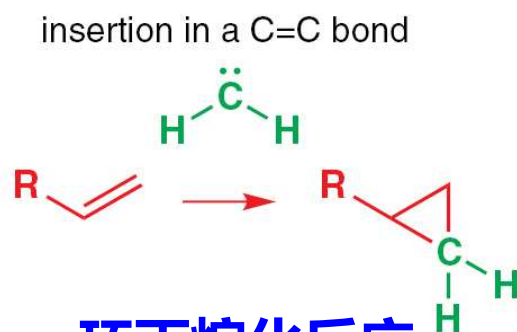
- Carbenes are **neutral** species containing **a carbon atom with only six valence electrons**.



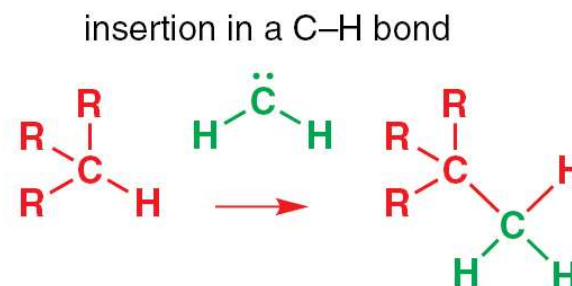
● Typical carbene reactions



氧氢插入反应



环丙烷化反应



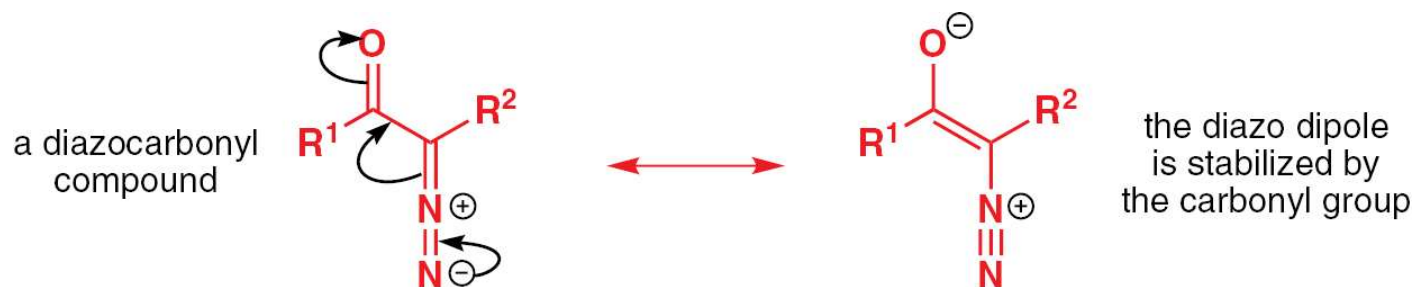
碳氢插入反应

The carbene inserts itself into a σ bond or a π bond.



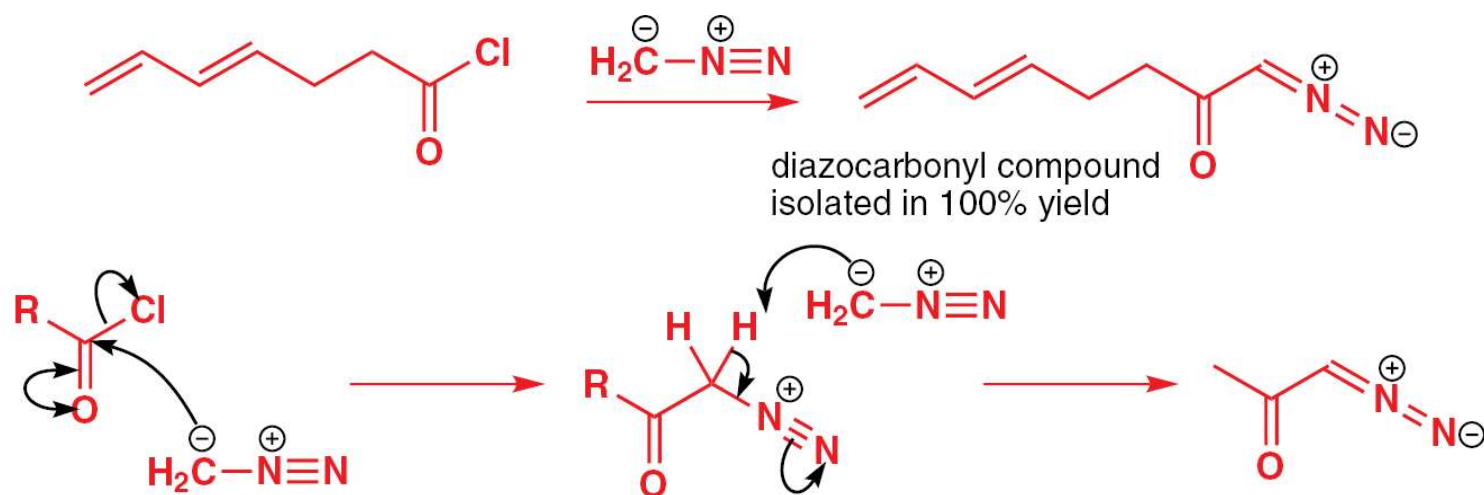
9.2 Ways to make carbenes

重氮化合物可作为卡宾的前体



重氮化合物的制备方法:

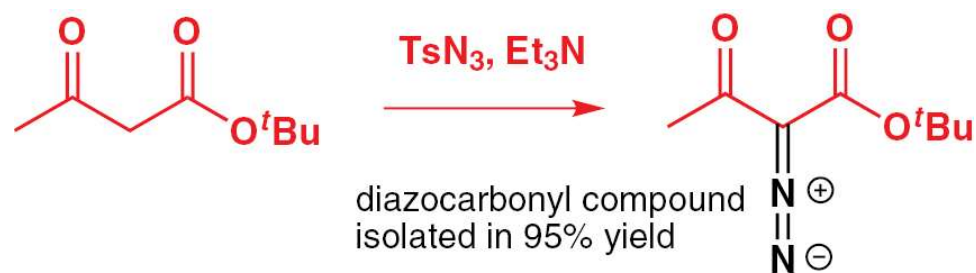
1. by reacting an acyl chloride with diazomethane





重氮化合物的制备方法

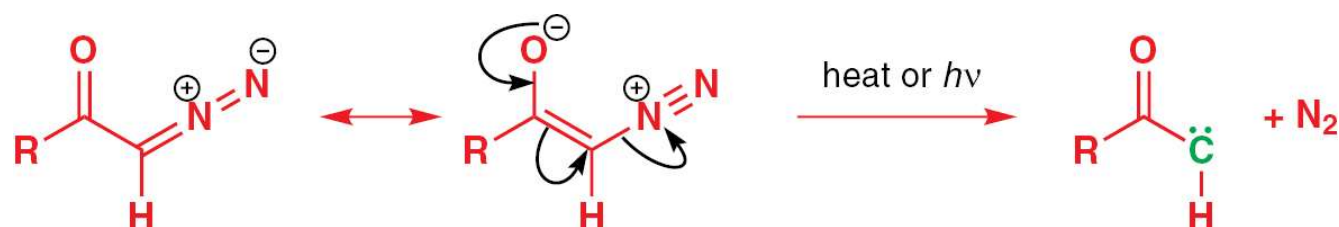
2. by reacting the parent carbonyl compound with tosyl azide, TsN_3 , in the presence of base.



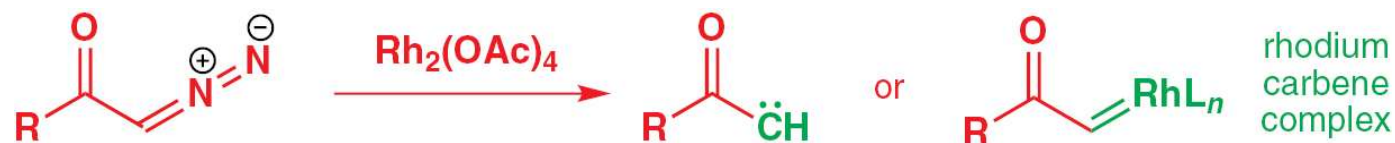


重氮化合物可作为卡宾的前体

Diazocarbonyl compounds can be decomposed to carbenes by **heat** or **light**.



Transition metal, such as **copper** or **rhodium** can promote formation of the carbene.

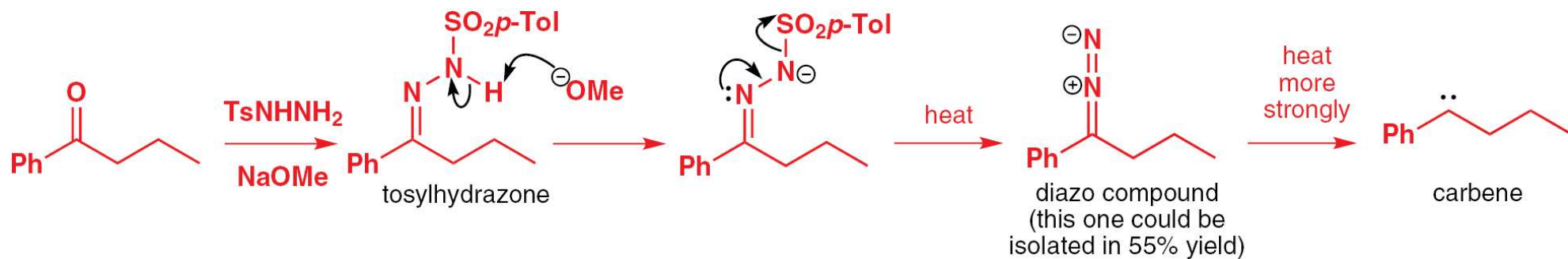


金属卡宾



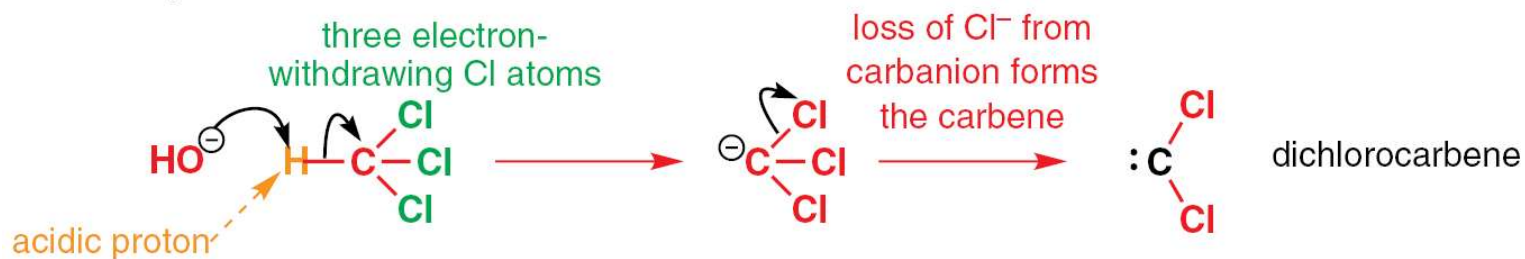
重氮化合物的制备方法

3. Carbenes from tosylhydrazones



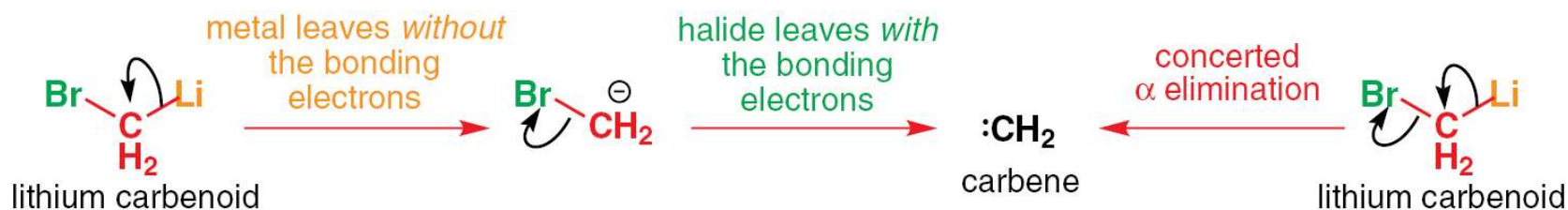
4. Carbene formation by α elimination

base-catalysed α elimination of HCl from chloroform





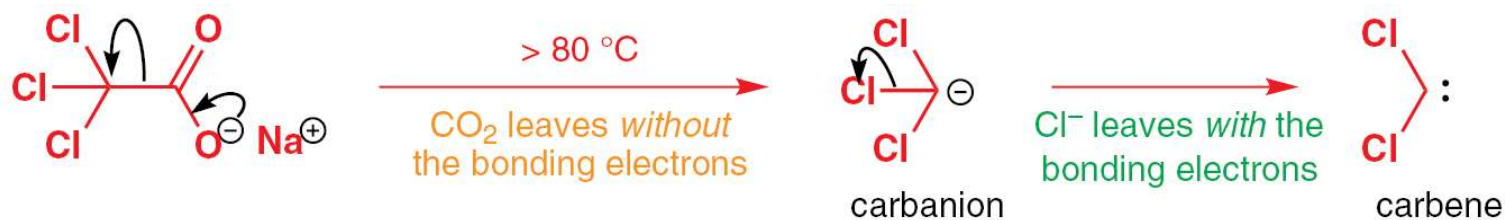
Carbene formation by α elimination



锂卡宾体



锌卡宾体

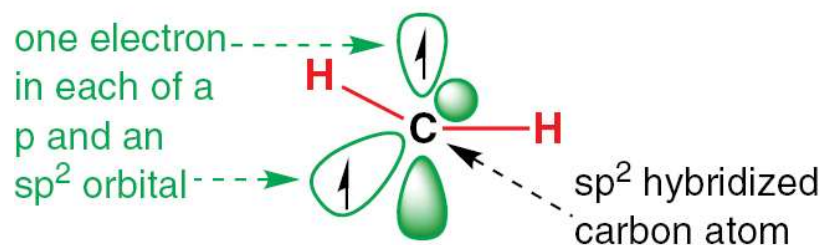




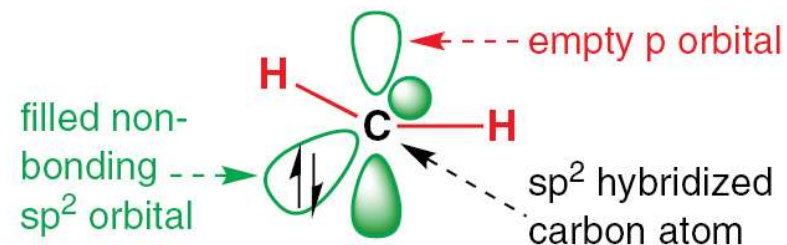
9.3 Carbenes can be divided into two types

Singlet and triplet carbenes (单线态和三线态卡宾)

two ways of arranging the electrons in a bent (sp^2) carbene:



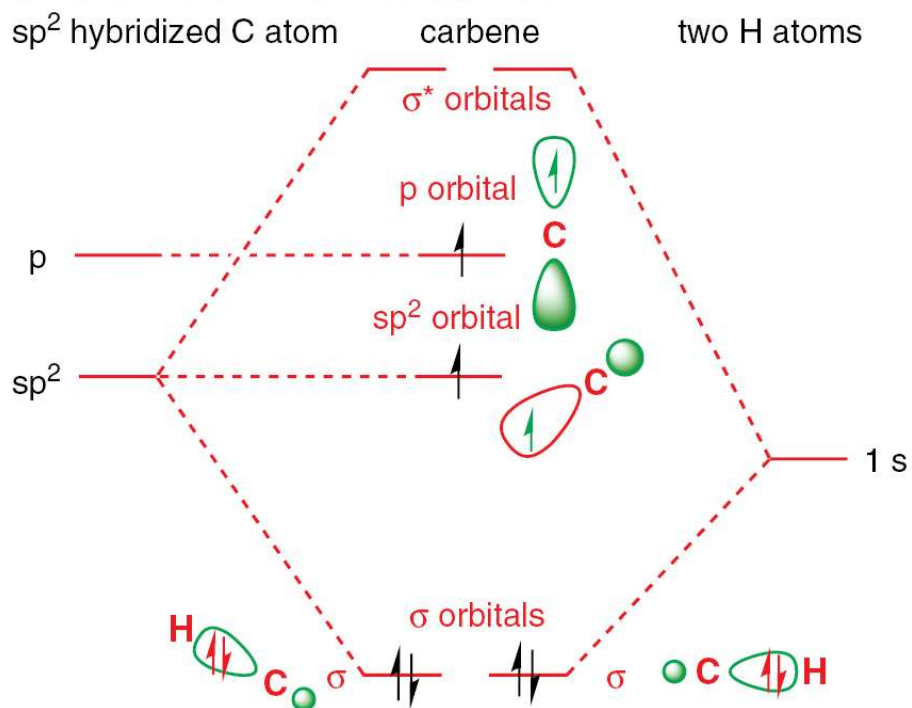
三线态卡宾



单线态卡宾

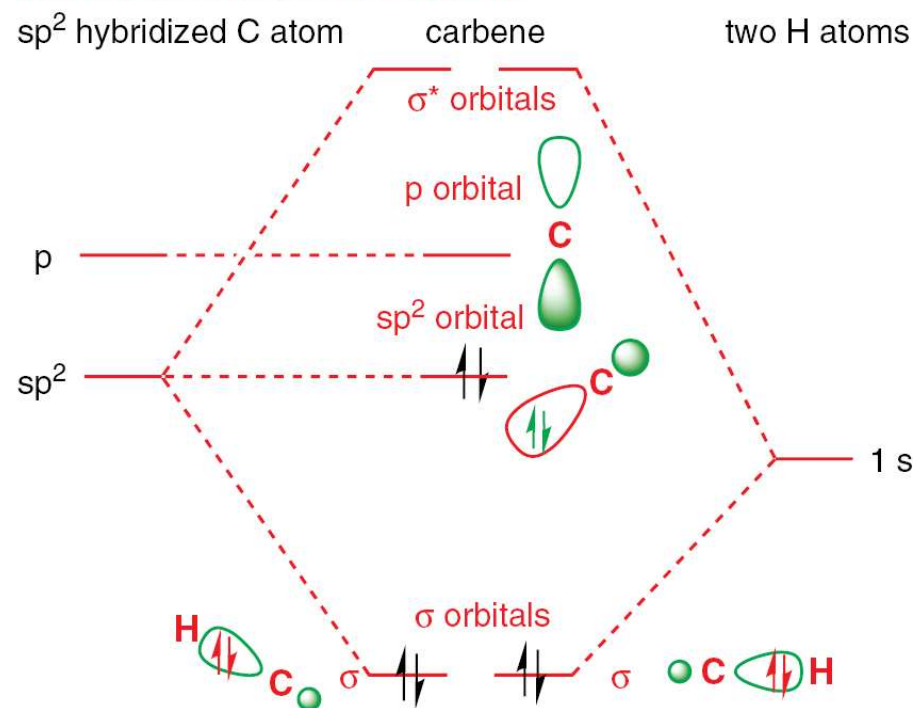


electronic structure of a bent (sp^2) triplet carbene with two unpaired electrons



三线态卡宾

electronic structure of a bent (sp^2) singlet carbene with two paired electrons

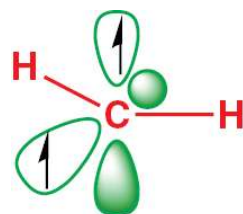


单线态卡宾

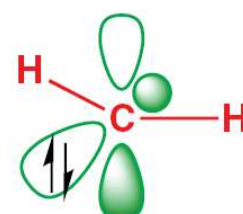


单线态和三线态卡宾的键角

三线态卡宾



triplet carbene

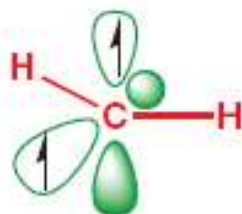


singlet carbene

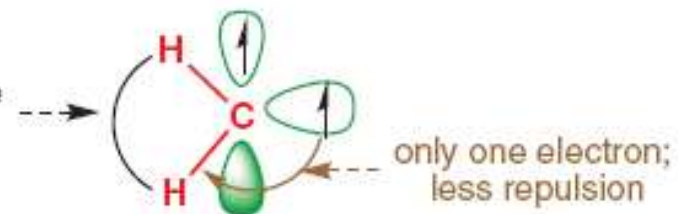
单线态卡宾

Triplet carbenes

two unpaired electrons:
observable by ESR



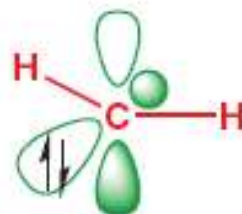
larger bond angle
(130–150°)



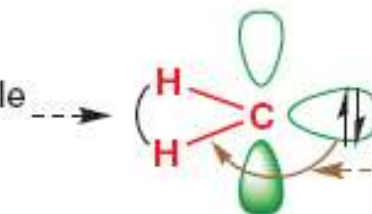
only one electron;
less repulsion

Singlet carbenes

no unpaired electrons:
not observable by ESR



smaller bond angle
(100–110°)



two electrons;
more repulsion



常见的单线态和三线态卡宾

Type 1: triplet carbenes

bond angle 130–150°

observable by ESR

:CH₂

:CHPh

:CHR

:CPh₂

Type 2: singlet carbenes

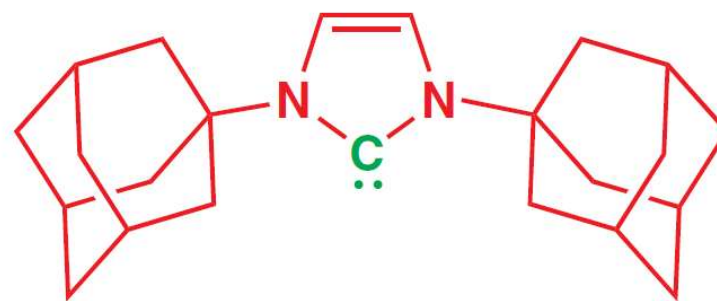
bond angle 100–110°

all electrons paired

:CCl₂

:CHCl

:C(OMe)₂



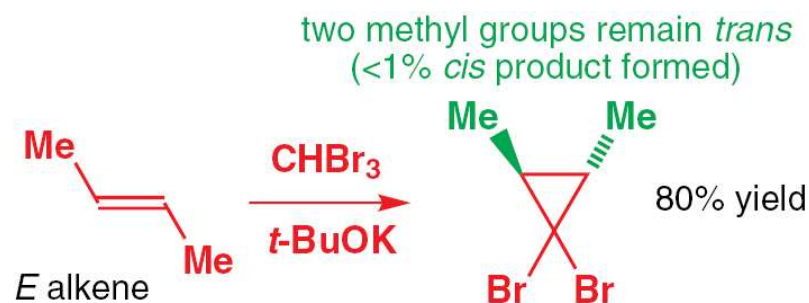
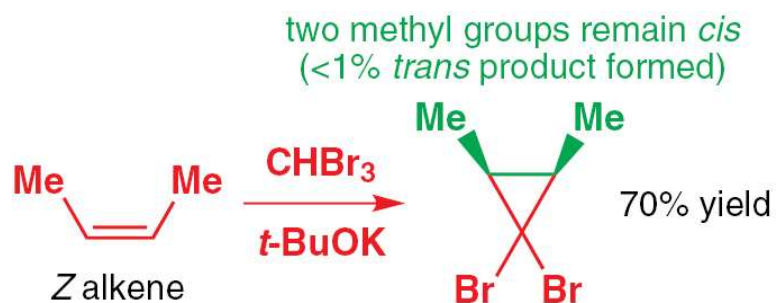
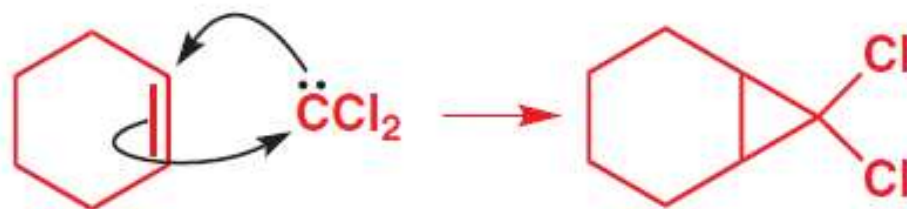


9.4 Carbenes react with alkenes to give cyclopropanes

Cyclopropanation of singlet carbenes

addition of singlet carbenes is a [1+2] cycloaddition

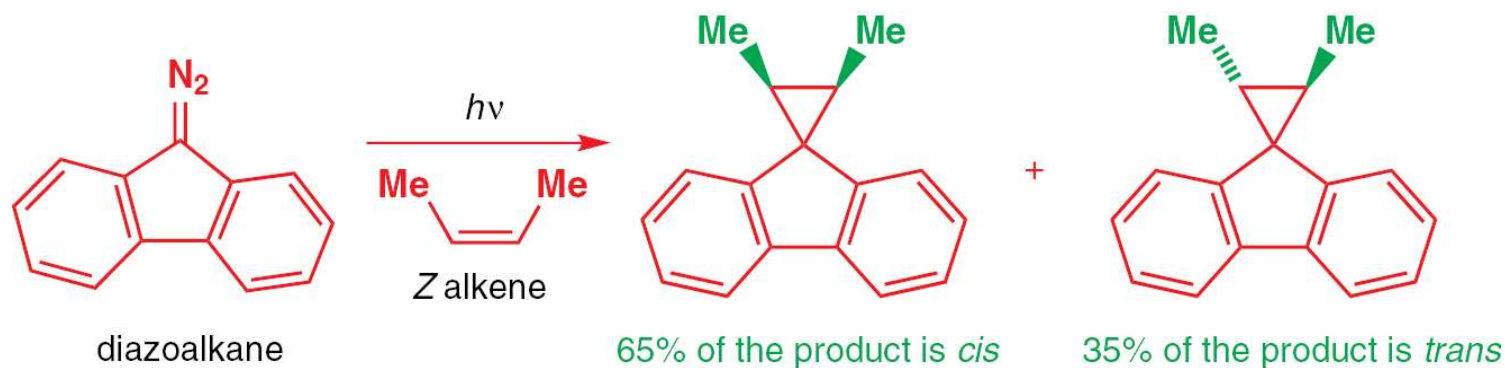
单线态卡宾协同加成



构型传递，立体专一

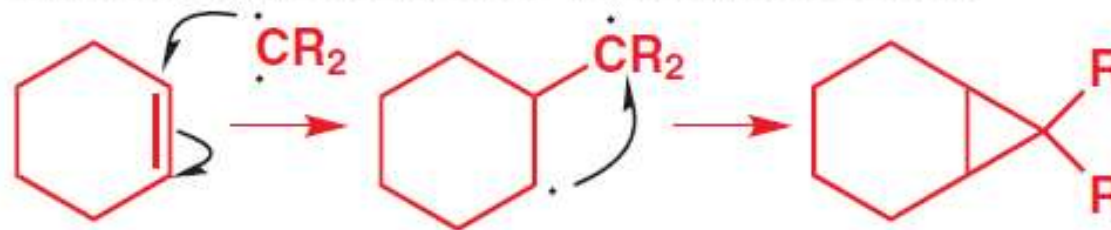


Cyclopropanation of triplet carbenes



构型不保持，立体不专一

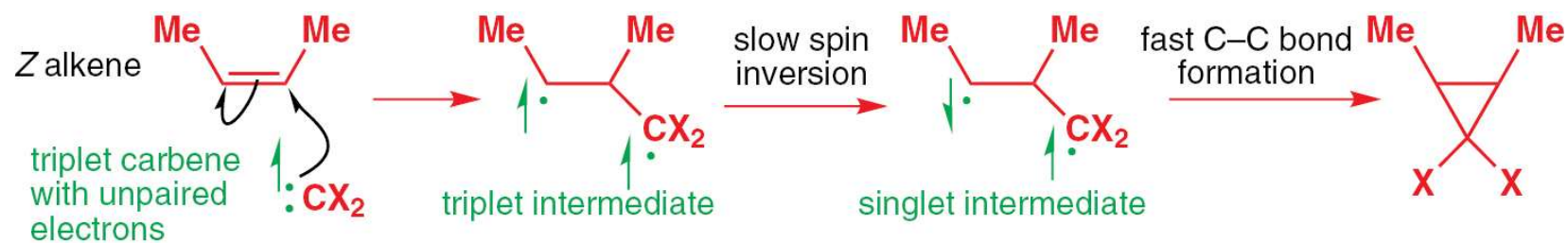
addition of triplet carbenes is a radical reaction



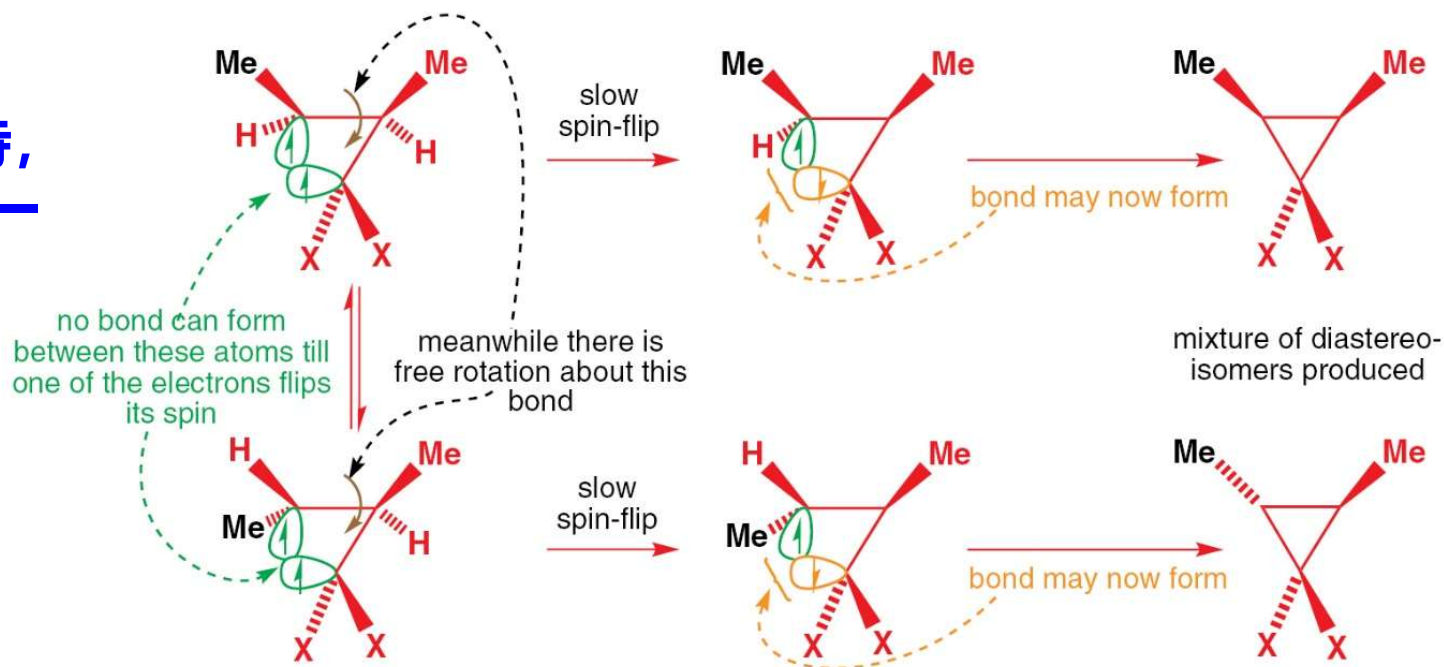
三线态卡宾：双自由基



Cyclopropanation of triplet carbenes



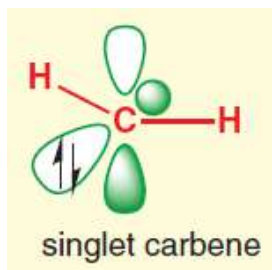
构型不保持，
立体不专一



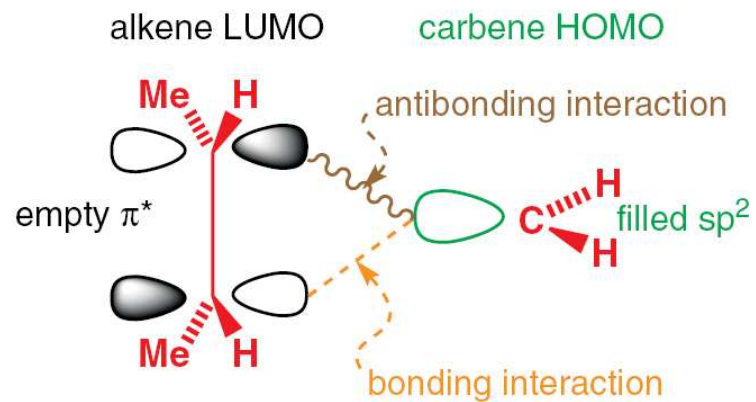
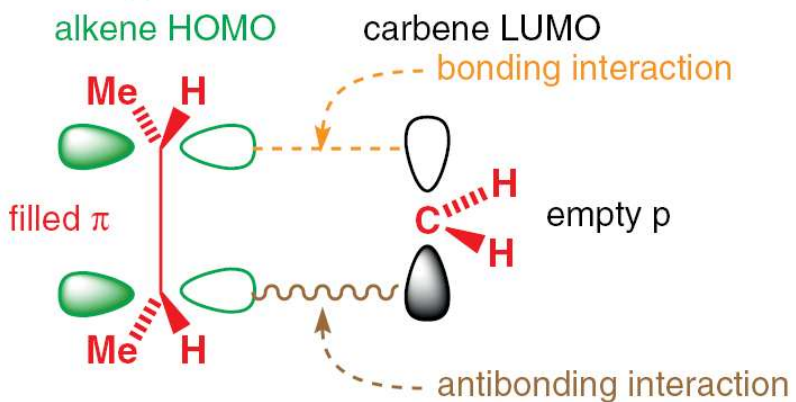


单线态卡宾的环加成反应遵循对称性守恒原则

Not allowed

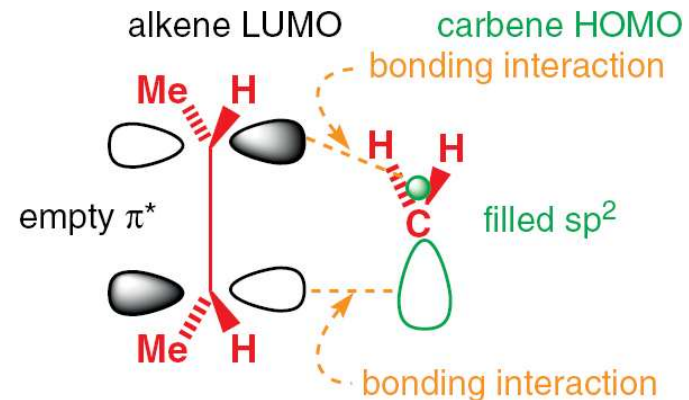
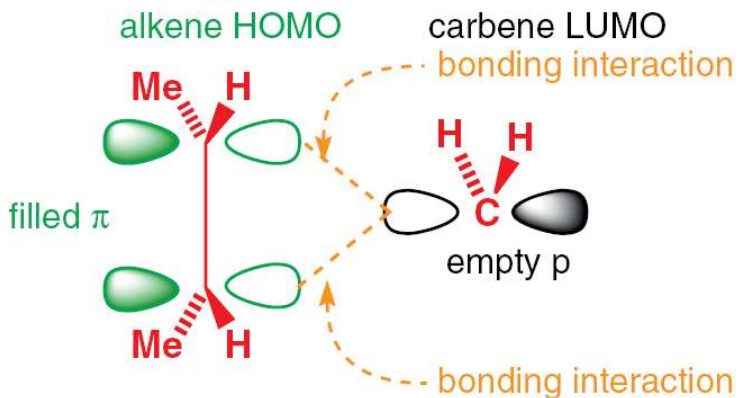


direct approach of carbene



Allowed

'sideways' approach of carbene

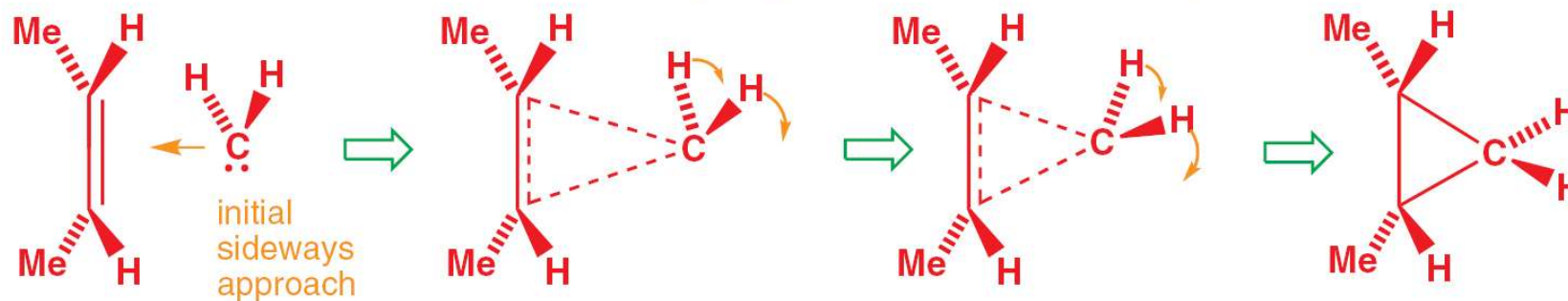




单线态卡宾的环丙烷化反应历程

'docking' of the carbene on to the alkene

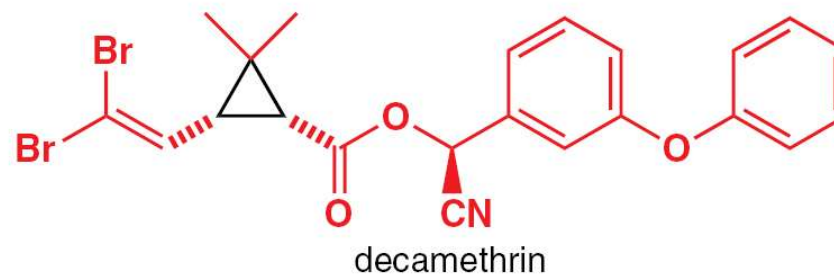
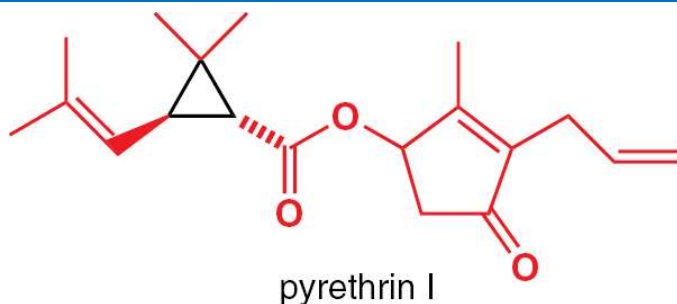
hydrogen atoms swing round into place



构型传递，立体专一



Making cyclopropanes



Simmons–Smith reaction

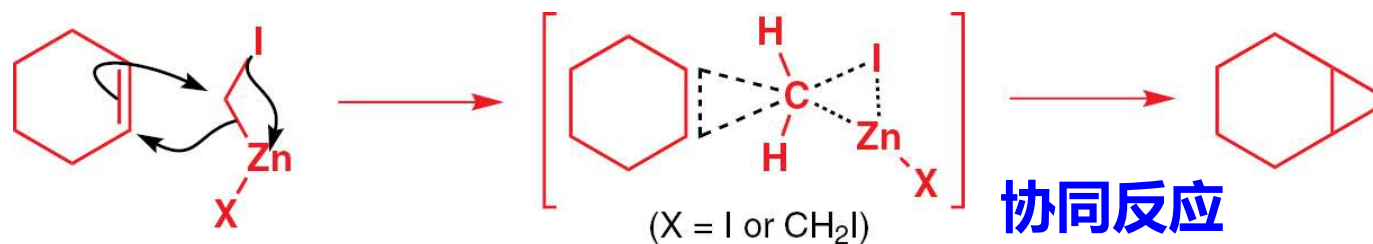
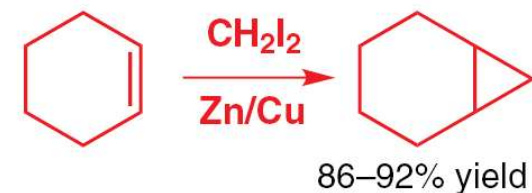
the Simmons–Smith reaction

formation of the zinc carbenoid

锌卡宾体

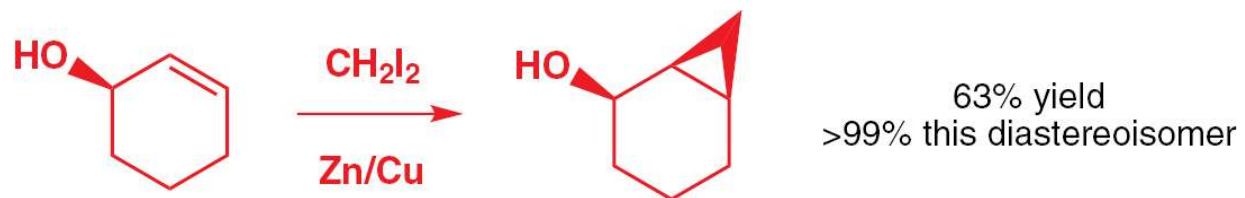


reaction of the zinc carbenoid

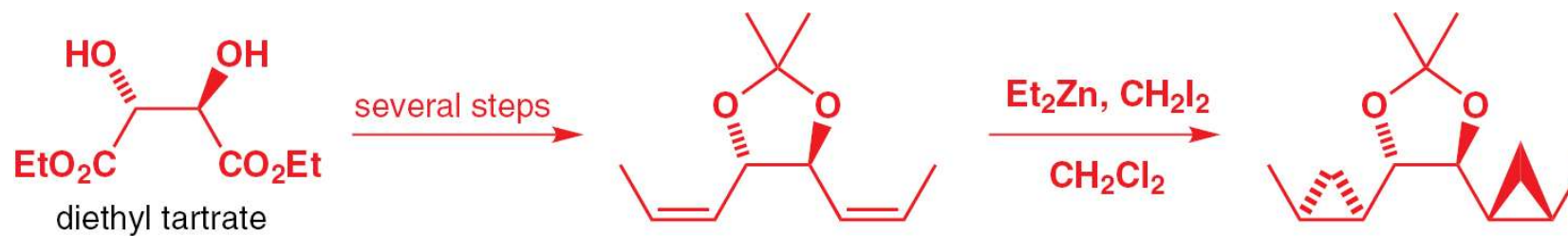




立体化学诱导 (配位导向)

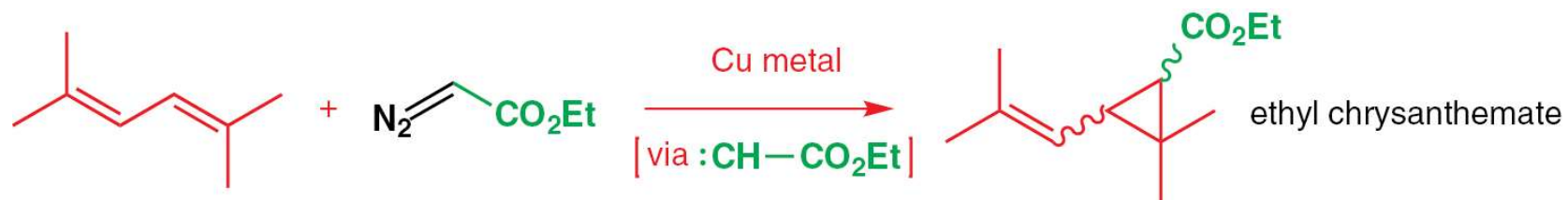


Stereochemistry

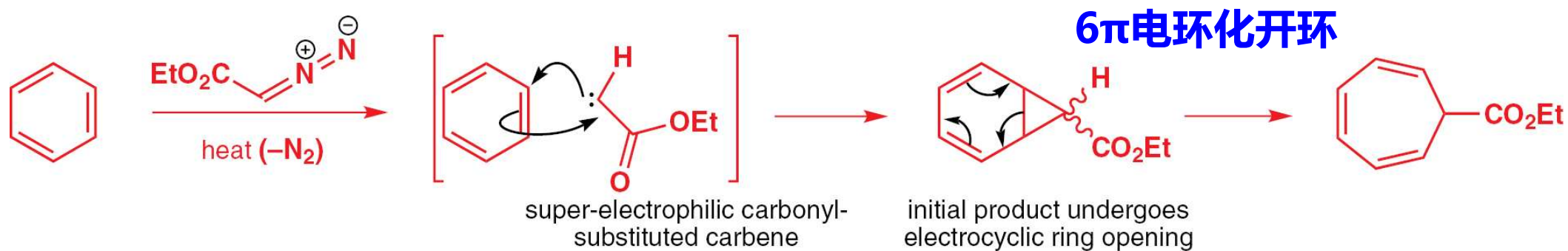




金属卡宾体



Buchner reaction

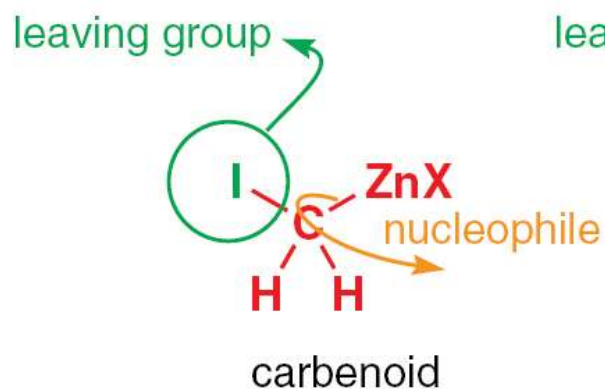




Common features for reagents for making three-membered rings

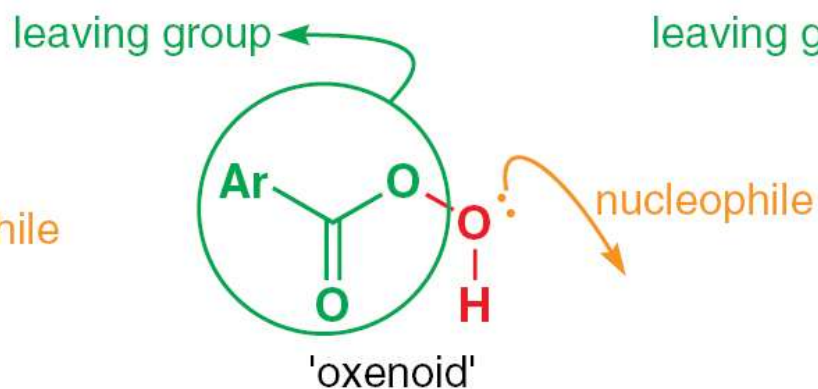
相似的构筑三元环

Simmons-Smith reagent



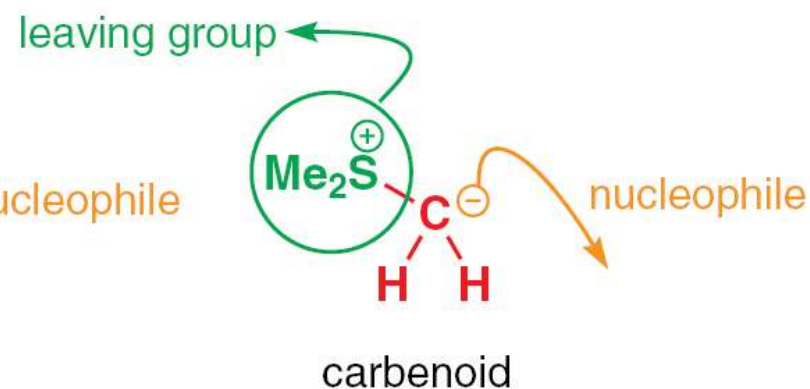
协同

peroxyacid



协同

sulfonium ylid

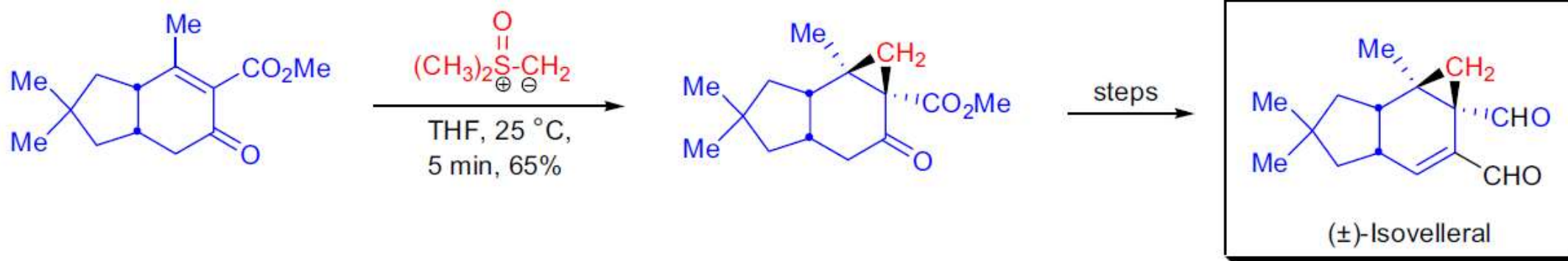
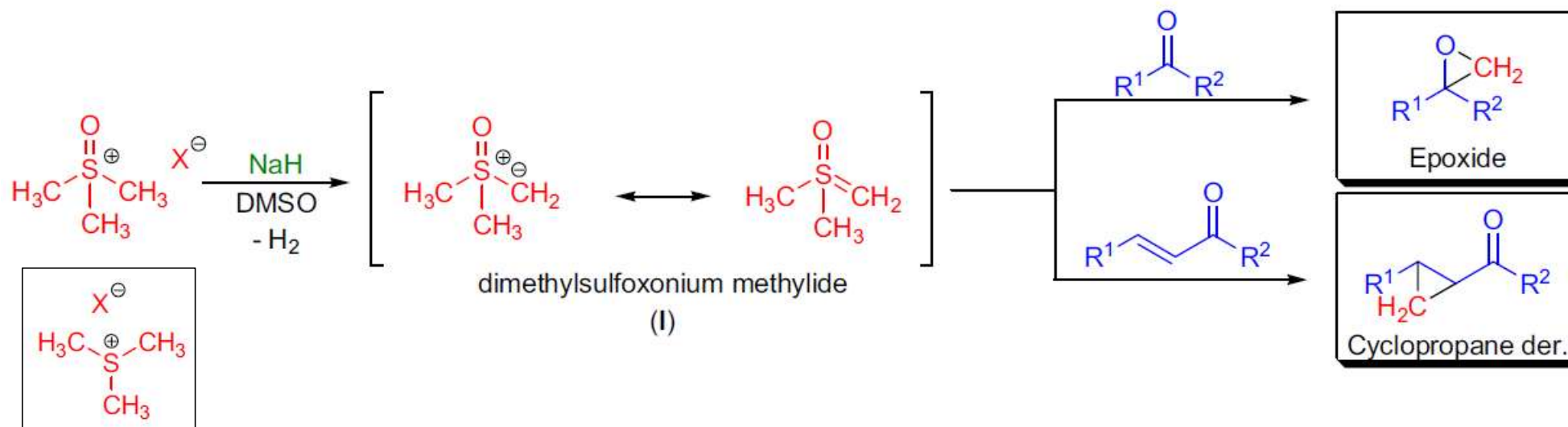


非协同

同时具有亲核试剂和离去基团：类卡宾的性质

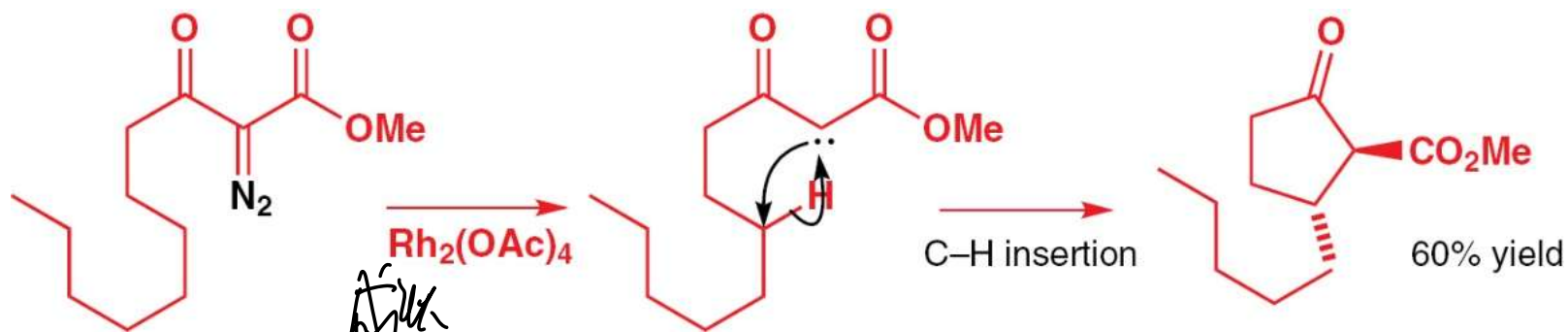
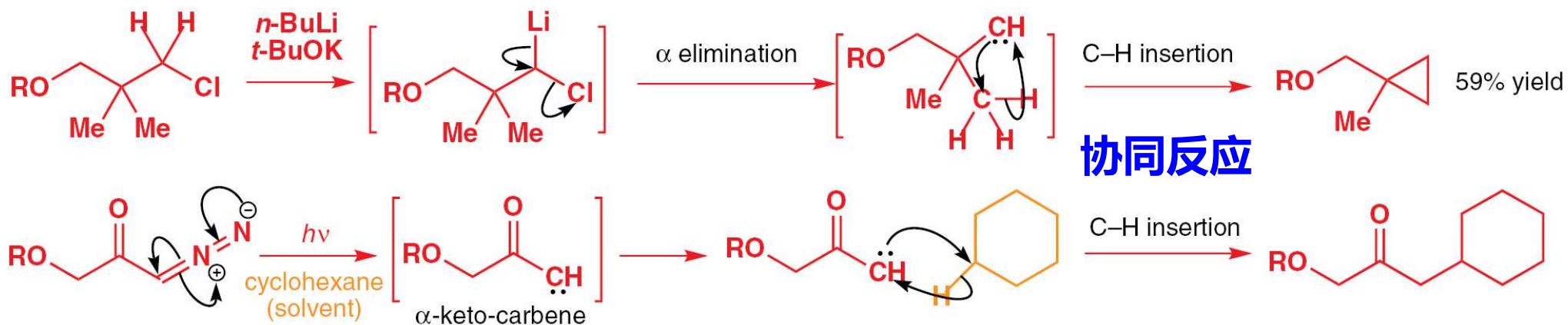


Corey-Chaykovsky Epoxidation and Cyclopropanation





9.5 Insertion into C–H bonds (碳氢插入反应)

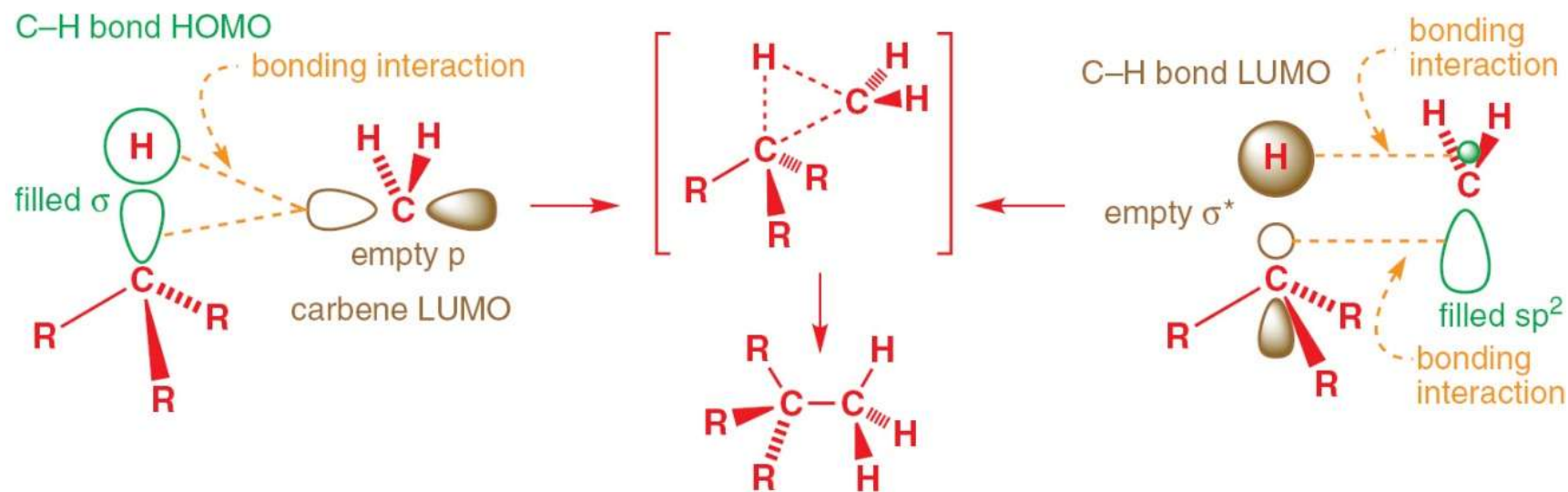


金属卡宾参与的碳氢插入反应 (优先形成五元环)



卡宾碳氢插入反应的分子轨道相互作用

orbital interactions during the insertion of a singlet carbene into a C-H bond



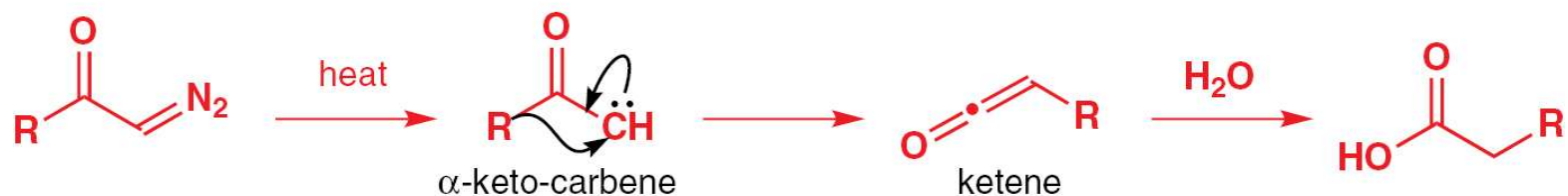
协同插入反应，构型保持



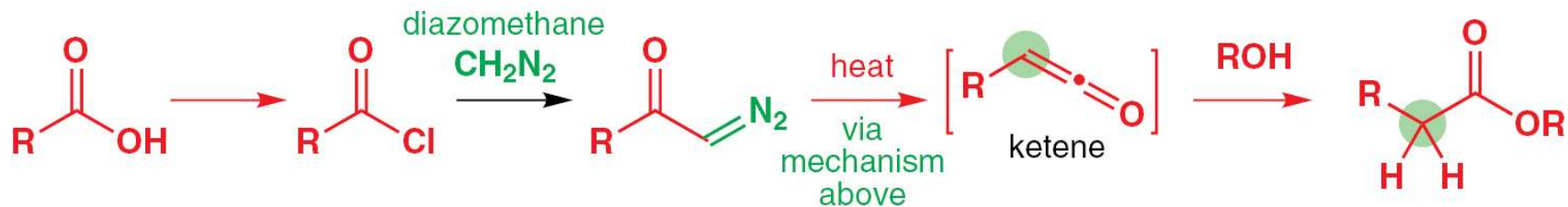
9.6 Rearrangement reactions

Wolff rearrangement

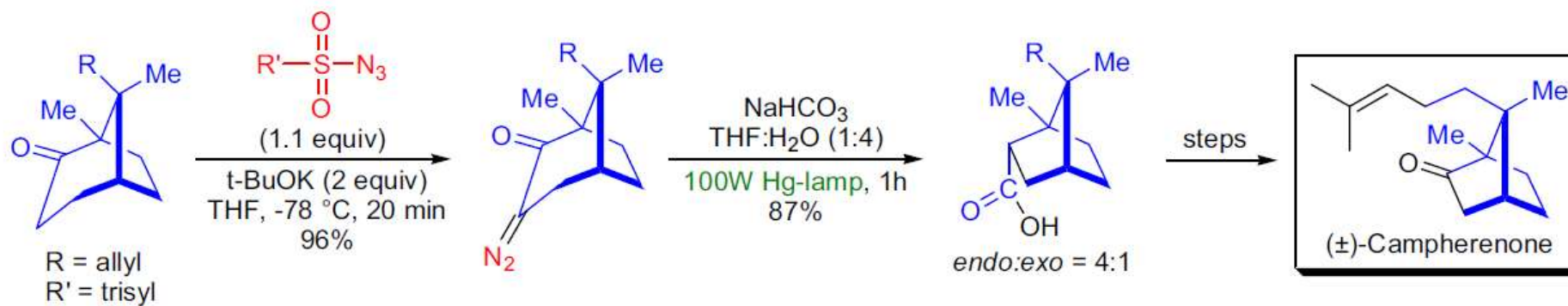
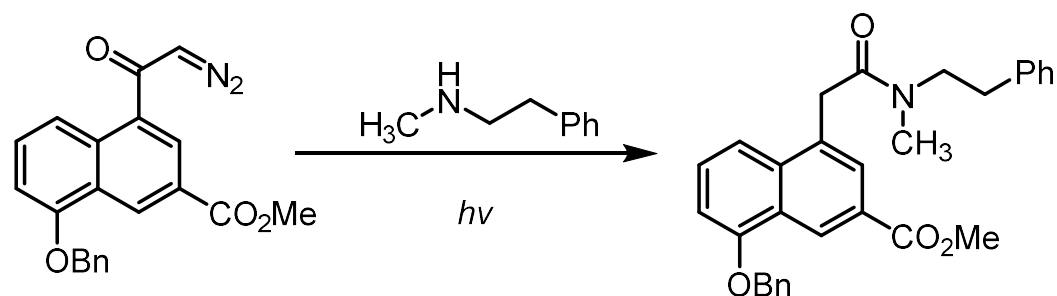
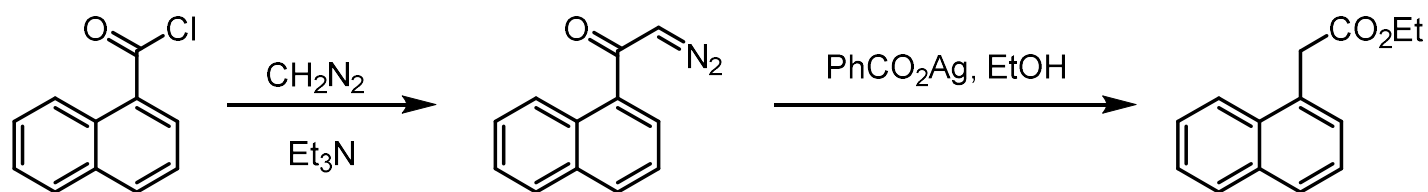
the Wolff rearrangement

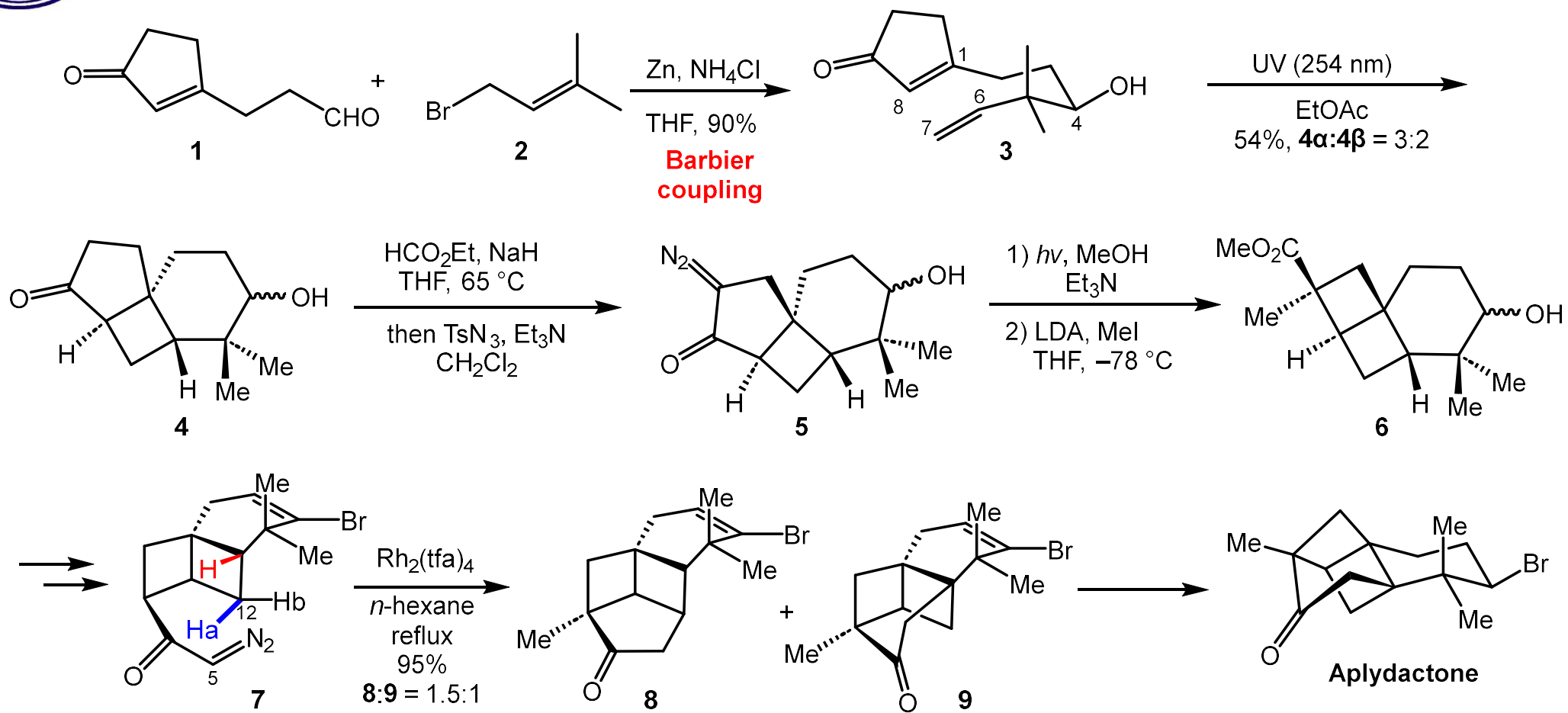


the Arndt-Eistert homologation



延伸一个碳，羧酸同系化



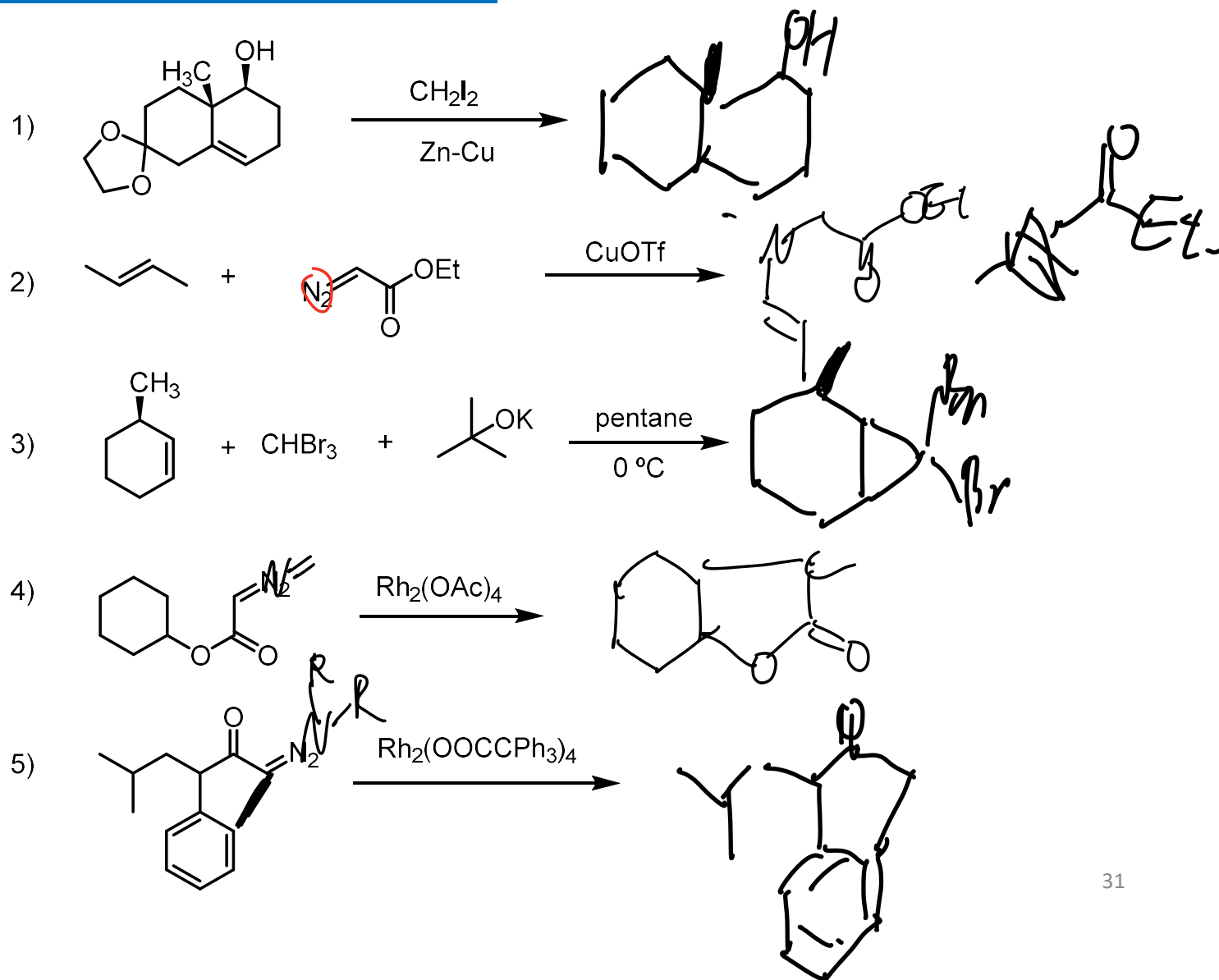


Angew. Chem. Int. Ed. **2017**, 56, 8187



练习

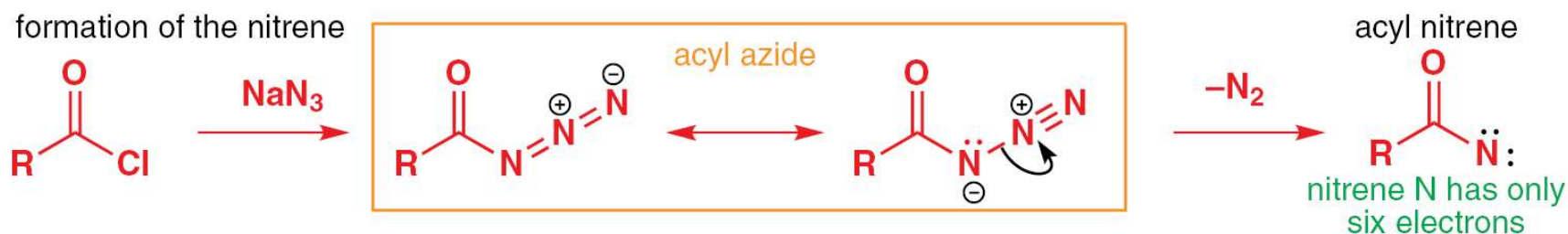
1. 写出下列反应的主产物



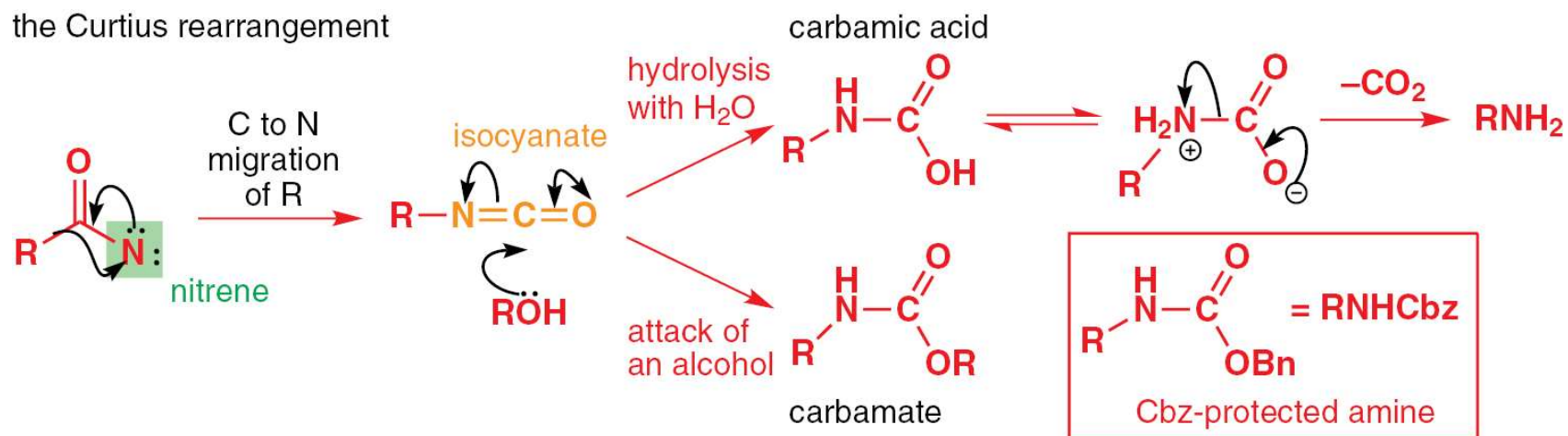


9.7 Nitrenes are the nitrogen analogues of carbenes

The nitrene N (乃春) has only one bond and has two lone pairs, making six electrons in all, like a carbene.



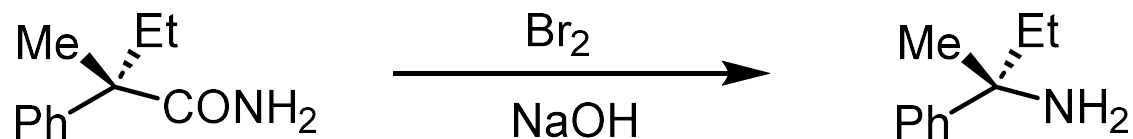
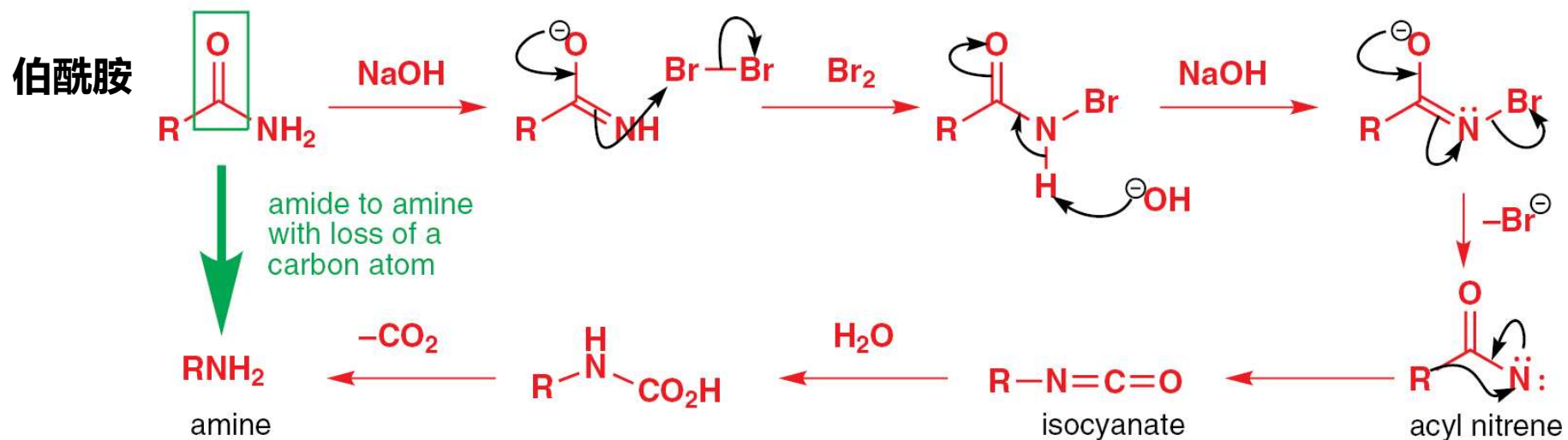
Curtius rearrangement





Hofmann rearrangement

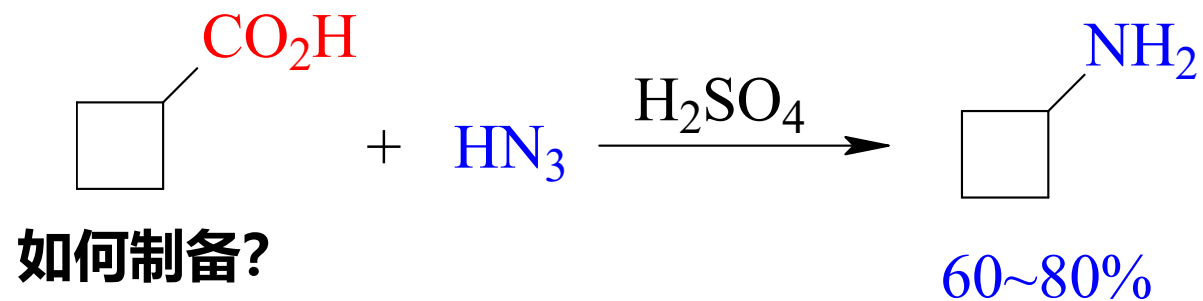
the Hofmann rearrangement



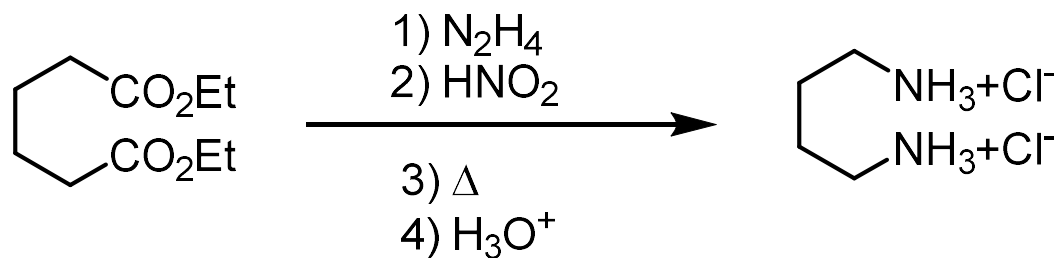
立体构型保持



Schmidt Rearrangement



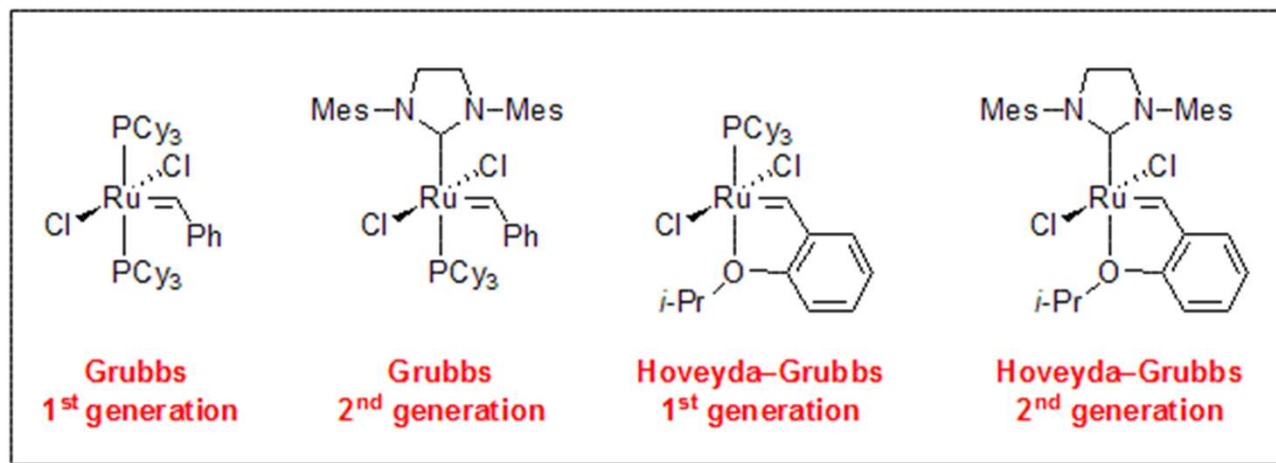
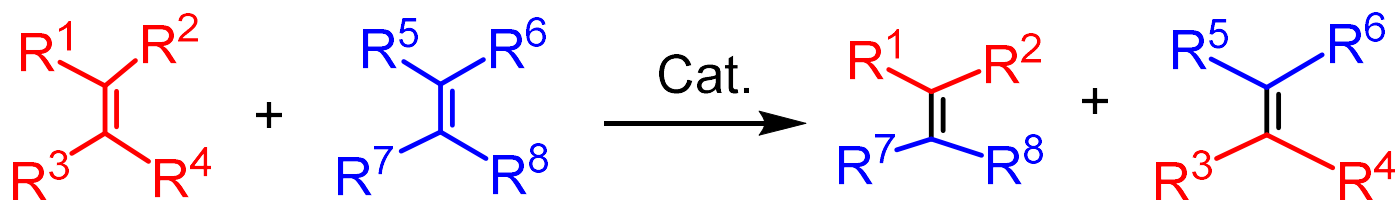
练习：画出下列反应的机理





9.8 Alkene metathesis

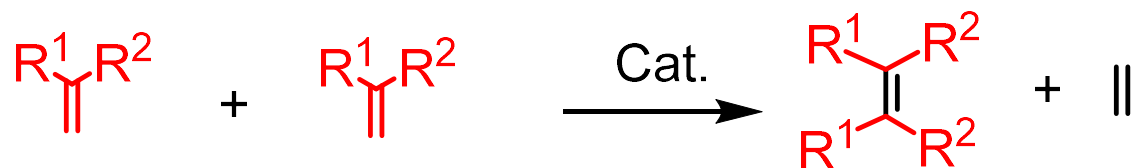
- Olefin metathesis is an organic reaction that entails the redistribution of fragments of alkenes (olefins) by the scission and regeneration of carbon-carbon double bonds.



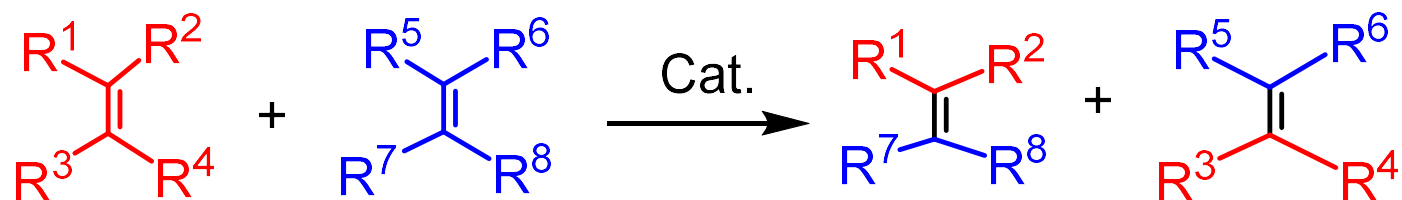


烯烃复分解反应的主要类型

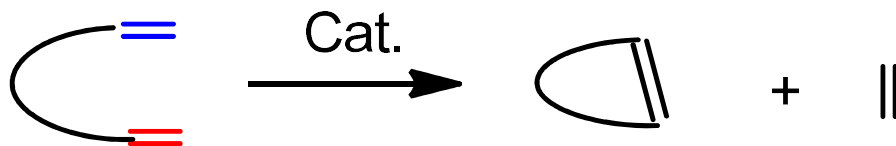
1. Self metathesis



2. Cross metathesis

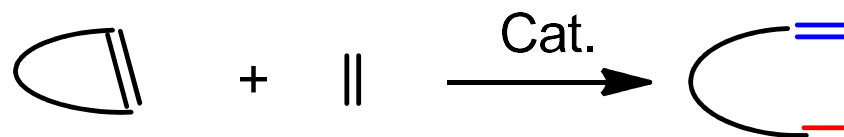


3. Ring-closing metathesis (RCM)

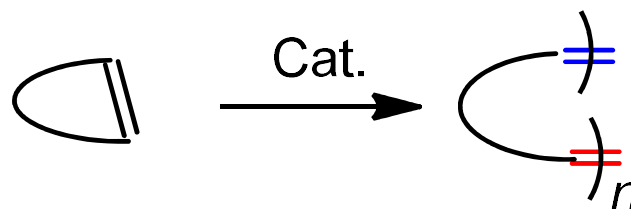




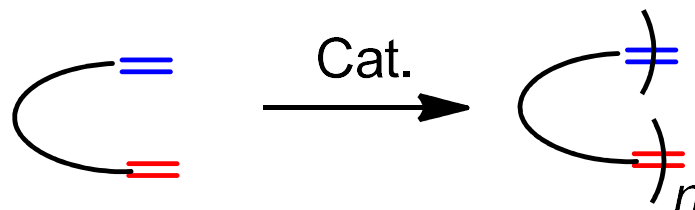
4. Ring-opening metathesis (ROM)

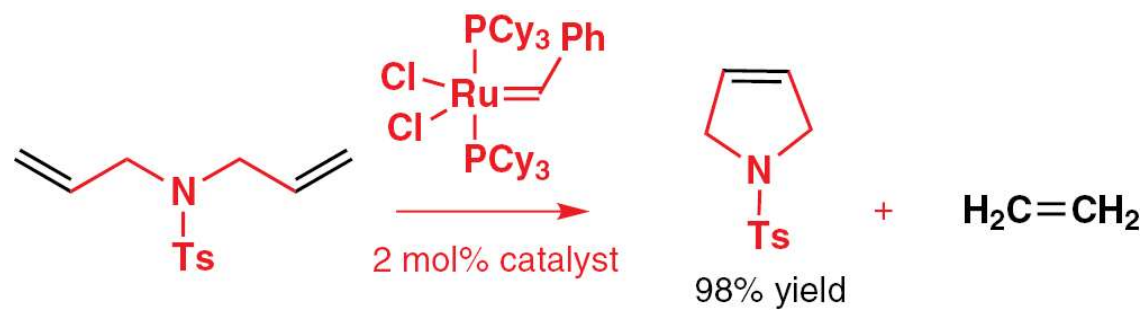
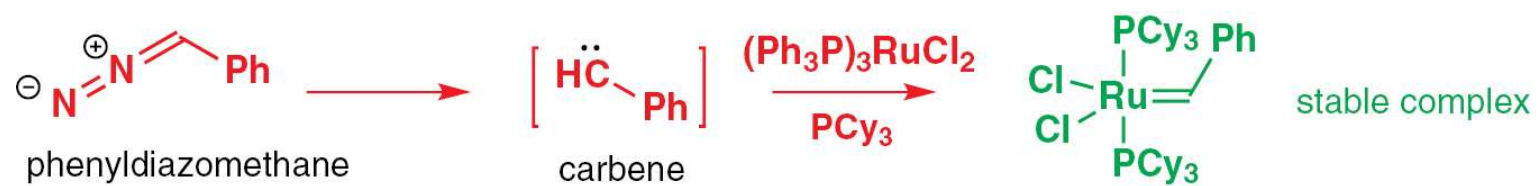


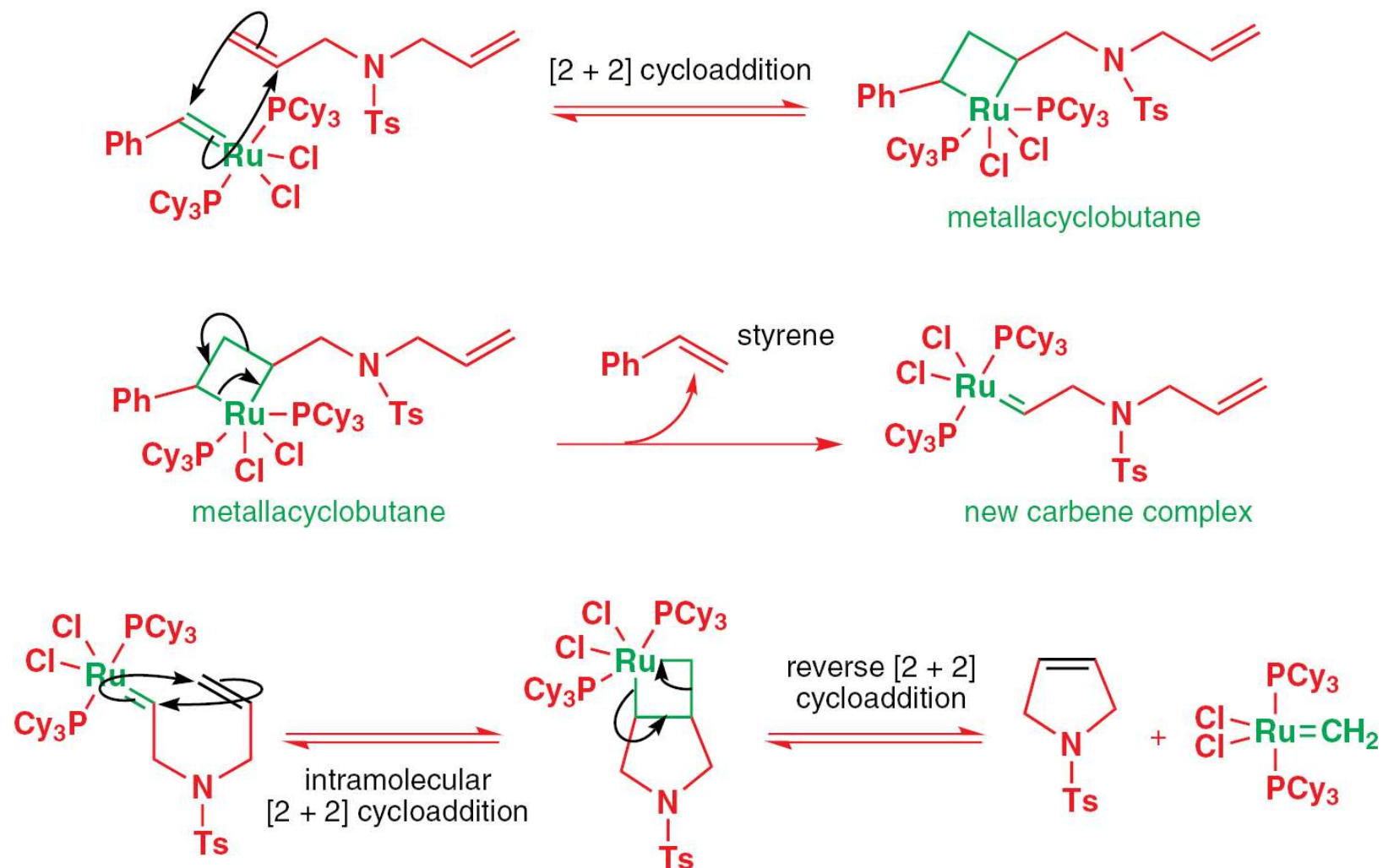
5. Ring-opening metathesis polymerization (ROMP)



6. Acyclic diene metathesis polymerization (ADMP)

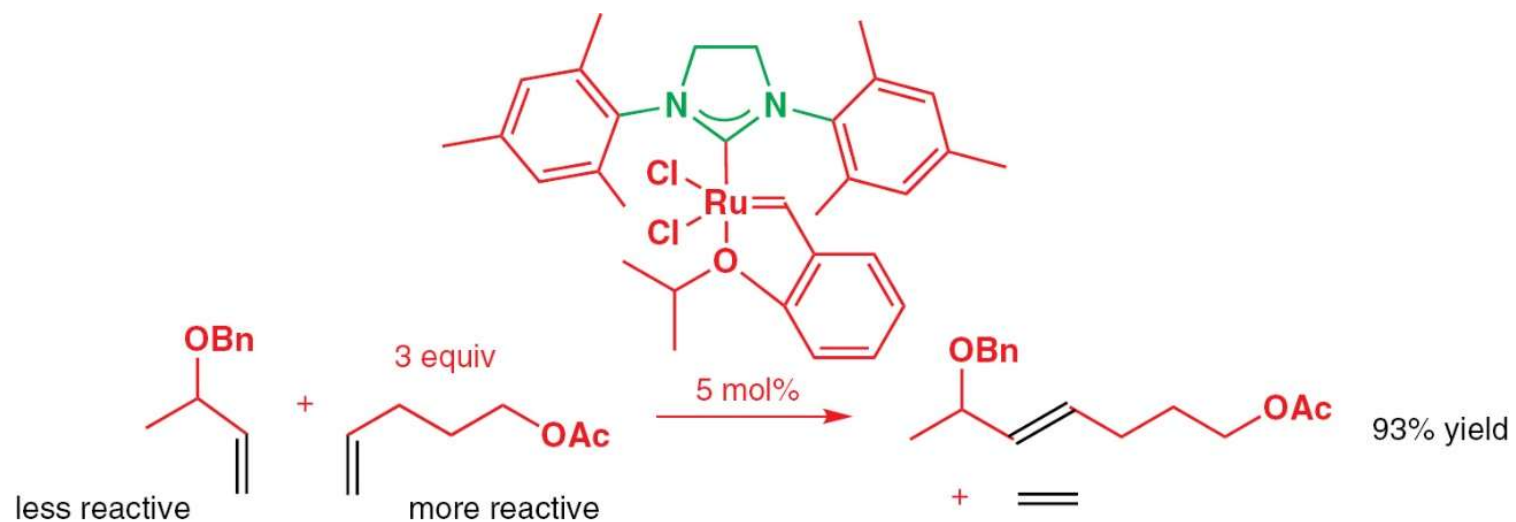








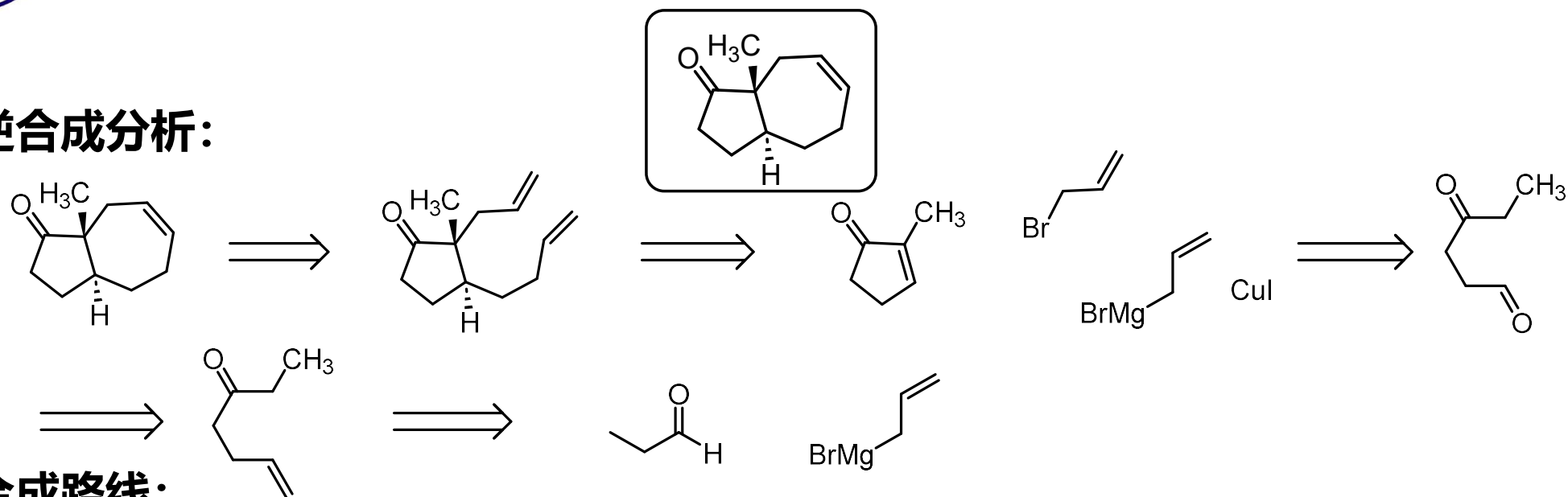
CM: 交叉烯烃复分解



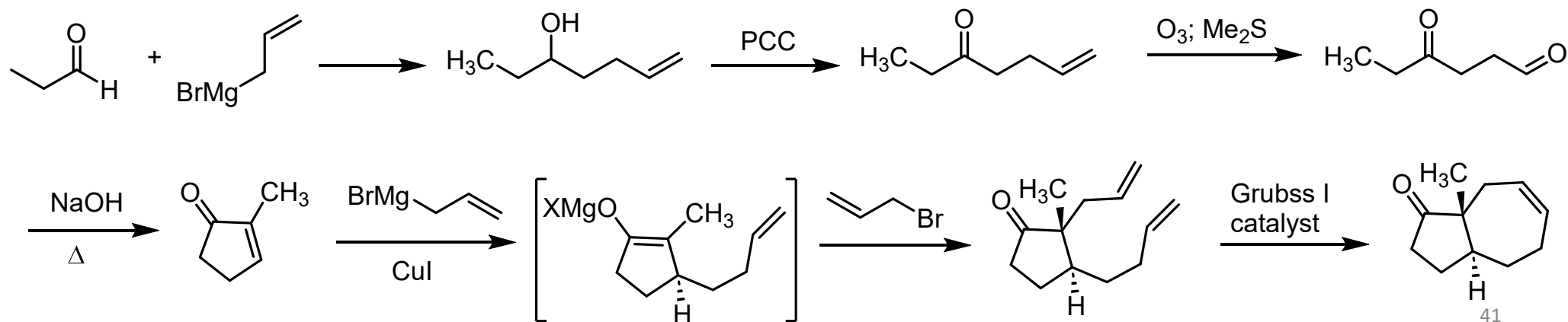


利用烯烃复分解反应合成下列化合物

逆合成分析:



合成路线:





9.9 Summary

1. 卡宾的类型：单线态、三线态
2. 常见卡宾（重氮化合物）的制备方法
3. 卡宾的反应类型：环丙烷化反应、碳氢插入、Wolff重排
4. 乃春的反应：Curtius重排, Hofmann重排
5. 烯炔复分解反应